

Engage Development Phase

Development Timeline: June 15 - July 15

Roles:

Arañez, Rocky B. - Fullstack Developer

Apidol, Ellaisa Faith Q. - Frontend Developer

Lim, Brandon M. - Backend Developer 1

Pineda, Joshua B. - Backend Developer 2

Sprint 0: Planning and Setup

June 15 – June 16

Goal

Prepare the project structure, database plan, API contracts, and task board.

Tasks

All Developers

- Finalize MVP scope.
- Create GitHub repositories.
- Set up Github
- Agree on coding standards.

Full-Stack Developer

- Create project structure.
- Prepare API contract document.
- Set up a shared .env guide.
- Set up branch workflow.

Backend Developers

- Set up Django REST Framework.
- Set up PostgreSQL.
- Create initial database models.
- Prepare authentication plan.

Frontend Developer

- Set up React Native project.
- Set up React web project.
- Create reusable UI components.
- Prepare mock data format

Output

- Repositories ready.
- Task board ready.
- API contract ready.
- Backend and frontend environments ready.

Sprint 1: Authentication, Users, and Base Dashboards

June 17 – June 22

Goal

Build the foundation for all role-based access.

Backend Tasks

Backend Developer 1

- Create custom User model.
- Add fields:
 - student_id
 - section
 - role
 - total_points
 - wallet_points
- Create login and registration APIs.
- Add token-based authentication.
- Add role-based permissions.

Backend Developer 2

- Create base models for:
 - Events
 - Rewards

- PointTransaction
- Feedbacks
- VolunteerTasks
- TaskApplication
- Create serializers.
- Create basic admin CRUD endpoints.

Frontend Tasks

Frontend Developer

- Build login and register screens.
- Build student mobile dashboard shell.
- Build instructor mobile dashboard shell.
- Build admin web dashboard layout.
- Build student web dashboard layout.

Full-Stack Developer

- Connect login/register to backend.
- Test role redirects:
 - Student
 - Instructor
 - Admin
- Review API responses.
- Fix integration issues.

Sprint 1 Output

- Users can register and log in.
- Role-based access works.
- Base dashboards are ready.
- Section field is included.

Sprint 2: Events and Geofence Attendance

June 23 – June 29

Goal

Build event management and mobile-only student attendance.

Backend Tasks

Backend Developer 1

- Create Event APIs:
 - create event
 - edit event
 - list events
 - view event details
 - close event
- Add geofence fields:
 - latitude
 - longitude
 - radius_meters
 - attendance_points
 - feedback_points
- Create AttendanceLog model and API.
- Create Heartbeat model and API.
- Implement server-side geofence validation.

Backend Developer 2

- Create offline attendance sync API.
- Add attendance statuses:
 - Pending
 - Verified
 - Rejected
 - Flagged
- Add duplicate attendance prevention.
- Add section-based attendance filtering.

Frontend Tasks

Frontend Developer

- Build admin Events screen.
- Build Create/Edit Event form.
- Build student mobile Events screen.
- Build student mobile Event Details screen.
- Add GPS capture in mobile app.

- Build student web event viewing only.

Full-Stack Developer

- Connect Event APIs.
- Connect geofence check-in API.
- Implement SQLite offline cache.
- Test offline-to-online sync.
- Verify student web cannot check in.

Sprint 2 Output

- Admin can manage events.
- Students can view events on mobile and web.
- Students can check in only through mobile.
- Offline attendance caching works.
- Synced logs can be verified by backend.

Sprint 3: Gamification, Rewards, Volunteers, and Instructor Features

June 30 – July 6

Goal

Build the engagement features and instructor role.

Backend Tasks

Backend Developer 1

- Create leaderboard API.
- Create student directory API.
- Create section attendance API.
- Create instructor section dashboard API.
- Create manual point awarding API.

Backend Developer 2

- Create reward catalog API.
- Create redemption API.
- Generate QR verification tokens.

- Create QR validation API.
- Create volunteer task APIs.
- Create volunteer application APIs.
- Add achievement unlock logic.

Frontend Tasks

Frontend Developer

- Build Rewards screen.
- Build Leaderboard screen.
- Build Profile screen.
- Build Volunteer application UI.
- Build Admin Students screen.
- Add “Award Points” button.
- Build Instructor mobile QR validation screen.
- Build Instructor web section attendance screen.

Full-Stack Developer

- Integrate points and leaderboard.
- Integrate reward redemption.
- Integrate QR verification.
- Integrate manual point awarding.
- Integrate volunteer approval workflow.

Sprint 3 Output

- Students can earn and view points.
- Leaderboard works.
- Rewards can be redeemed.
- QR verification works.
- Admin can award manual points.
- Instructors can view section attendance.
- Students can apply as volunteers.

Sprint 4: Feedback, Sentiment, Reports, Testing, and Deployment

July 7 – July 15

Goal

Finalize the MVP, test all modules, and prepare deployment/demo.

Backend Tasks

Backend Developer 1

- Finalize attendance validation.
- Finalize suspicious log flagging.
- Finalize section-based reports.
- Prepare DigitalOcean deployment settings.
- Secure environment variables.

Backend Developer 2

- Create feedback submission API.
- Add spam/not-spam label support.
- Add sentiment classification field.
- Prepare sentiment model endpoint.
- Create sentiment report API.
- Connect model output to feedback records.

Frontend Tasks

Frontend Developer

- Build feedback submission screen.
- Build admin sentiment reports screen.
- Build attendance report filters.
- Polish mobile UI.
- Polish web dashboard UI.
- Add loading, success, and error states.

Full-Stack Developer

- Deploy backend to DigitalOcean.
- Test PostgreSQL production database.
- Test Cloudinary uploads.
- Test API integration.
- Perform end-to-end testing.
- Prepare demo accounts and sample data.

Sprint 4 Output

- Feedback submission works.
- Sentiment reports are visible.
- Backend is deployed.
- Core workflows are tested.
- System is demo-ready.